Run report: e7b_k0

2026-10-18, 16:06:42 to 16:09:45 UTC. research/isogeny_conductor_gap_round5_20261012, lead E7b.

Spec specs/e7b_k0.json. **Session** ECBS1heff1e81e9e36 at commit 6574178b6 (git_dirty false), audited with `ecbench verify --replay 12`: ok, 80 of 80 verified, 12 replays reproduced. Isolation earned L2 on 44 records, L1 on 36. **Workloads:** the a = 0 Koblitz curves over $F_{2^{19}}$ and $F_{2^{23}}$ (h = 4), 4 targets × 2 rounds. **Arms:** rho.negation; blind F4 descent and blind MITM on binary-subspace:dimension=9 with and without trace_kernel=1.

curve	arm	relations / trials	F4 ops / relation	max degree	S	wall / run
n = 19	f4 default	808 / 3,488	4,329,947	3	4.93	2.93 s
n = 19	f4 trace-kernel	952 / 2,777	4,964,166	3	4.90	5.15 s
n = 23	f4 default	758 / 47,189	4,634,927	3	5.43	7.63 s
n = 23	f4 trace-kernel	929 / 29,782	3,329,596	3	3.86	6.74 s

Derived (TABLES_E7.md): yield trace-kernel ÷ default 1.48 [1.37, 1.60] (n = 19) and 1.94 [1.77, 2.14] (n = 23); F4 ops per relation default ÷ trace-kernel 0.87 and 1.39. MITM S 195.75 ⟶ 160.82 and 53.13 ⟶ 49.69; rho S 1.72 and 1.18. Wall times are host observations.

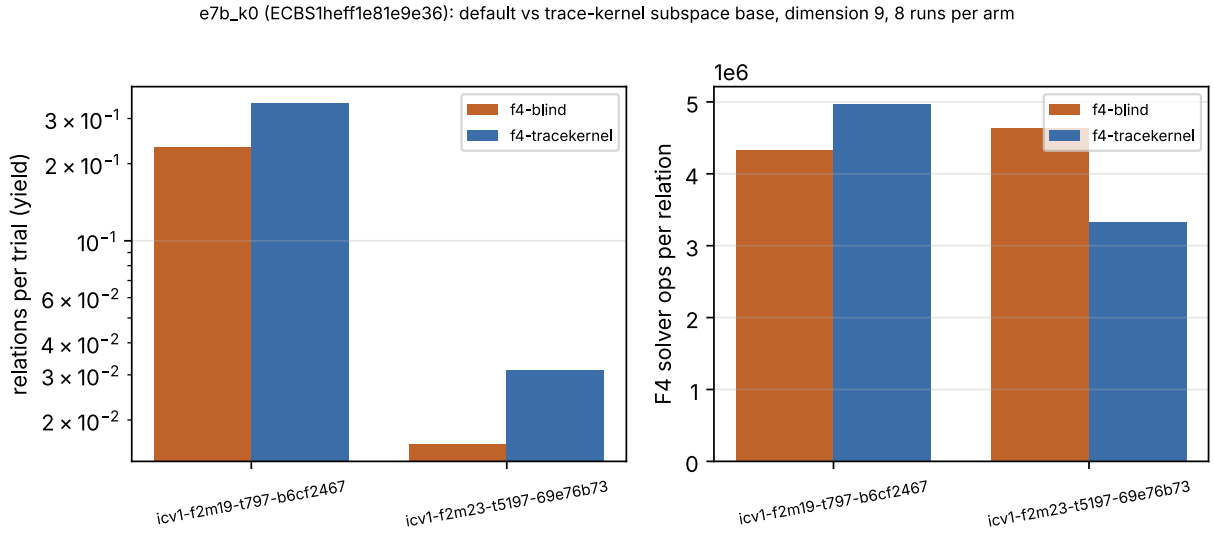

Figure 1: Session ECBS1heff1e81e9e36: relations per trial (log scale) and F4 solver operations per relation, default against trace-kernel base.